

**Subject: RE: Multi-Grid TBox Network — Routing, Virtual Circuits and Adaptive COSTK (Rev.7)  
— Review and Sequencing**

**Hi Wulun,**

Thank you for the updated Rev.7 document. Your overall architecture is a strong piece of work — the three-layer separation you've formalized (EasyMesh owning radio/backhaul link management only, Babel owning ordinary IP reachability, and the new GVC controller/agent owning explicit MPLS circuits for admitted traffic) is exactly the structure we converged on independently in our own Section 9 analysis in late August. Seeing it worked out in this level of detail, with explicit non-overlap rules between the three layers, is a good sign that the direction is sound.

After we (including you) met with Tachyon last week, we have sent them the NDA. We've decided to finalize the Relay T-Box development schedule with Tachyon first, and only bring Grid T-Box development forward once that is settled. As part of that same sequencing, our MaxLinear technical inquiry — which includes the hardware/firmware question this document depends on most (can EasyMesh retain radio-level control of DFS/onboarding while exposing a routable interface to Babel on the same radio) — will also go out only after the Relay T-Box development agreement with Tachyon is in place.

That means this review of your Rev.7 design is groundwork for Grid T-Box's L3 routing architecture, not something we're asking you to build against immediately. Please treat our comments below in that light — as input for refining the design and the vendor question list, rather than an implementation go-ahead.

**On EasyMesh control-domain assumptions**

Your Rev.7 explicitly moves away from the two-zone overlapping partition (Controller A/B, row-3 dual-homed) we described in our Integration Request, calling it a vendor-qualification scenario rather than the selected scalability guarantee, and proposes bounded, non-overlapping EasyMesh control cells instead. We don't disagree with treating this as open until tested, but we'd like your explicit confirmation: is the bounded-cell approach now your recommended baseline, superseding the two-zone plan, or are both still candidates pending Phase 0 vendor data? Either way, please note that the actual answer — how many Agents one Controller can manage on our target silicon — is a MaxLinear question (our own inquiry item 5a) that has not been asked yet given the sequencing above. We'd like your E01–E10 test plan to stay written so it can validate either topology once that hardware answer comes back, rather than committing to one now.

**On cross-grid redundancy — a concrete proposal for the circuit model (N+1): please see the separate attachment**

Related to the point above, we'd like to propose a specific answer to the bounded-cell-vs-two-zone question, and at the same time close the Relay-redundancy gap you flagged in your August 30 proposal (a single long-haul relay pair being a single point of failure between grids). The idea: each grid keeps its own primary uplink relay (Relay A for Grid A → Govnet A, Relay B for Grid B → Govnet B),

and we add one more relay pair — Redundant A and Redundant B, both ordinary Relay T-Boxes — linked directly to each other within relay range, sitting between the two grids. That one cross-link does two jobs: it reroutes Grid A's traffic out through Govnet B (and vice versa) if either Govnet uplink fails, and in normal operation it carries overflow traffic from whichever grid is over capacity to the one with headroom.

We think this maps cleanly onto two GVC CircuitIntents, mode=PROTECTED, one per grid uplink — each with a primary\_path over its own Relay and a backup\_path routed through Redundant A/B and the other grid's Relay, tagged diversity=shared\_risk. Just as importantly, it means Grid A and Grid B can each stay a single independent bounded EasyMesh cell: the cross-grid redundancy lives entirely at the GVC/Babel layer, so it never requires merging or splitting an EasyMesh controller domain.

One thing we'd like your read on specifically: your own Section 8.1 rule against per-packet ECMP means a PROTECTED backup\_path is dormant until the primary fails, whereas what we want in normal operation is closer to continuous load sharing between the two grids. We see two ways to reconcile that — (a) issuing a second SOFT/BEST\_EFFORT circuit on the same physical Redundant A↔B link, dedicated to overflow traffic, or (b) handling it purely through admission policy, routing new admissions to the other grid whenever the local capacity ledger is out of headroom, with no mid-session path change. We'd like your opinion on which fits the GVC model better, or whether there's a third pattern you'd already assumed. A diagram and a short slide deck laying this out in full are attached — the underlying assumptions are that Grid A and Grid B sit within relay range of each other, that both are served by the same carrier (DICT), and that at any given time at least one of the two has spare capacity.

### **On MPLS support on the target firmware**

We noted that Section 7.1 already flags the captured Edge TBox kernel configuration as missing CONFIG\_MPLS\_ROUTING and CONFIG\_LWTUNNEL, meaning native MPLS forwarding requires a full firmware rebuild on the exact vendor tree, not a module add-on. We read this as new engineering scope rather than a gap in your design, and we agree with treating it as a Phase 0 gate. Since this rebuild depends on MaxLinear's toolchain and kernel tree, it will also need to wait for our MaxLinear engagement to formally open — we'd like to fold this kernel-capability question into that same inquiry rather than have your team chase it independently in the meantime.

### **On mobility requirements (we will discuss it later)**

Thank you again for the thorough work on this revision. Happy to set up a call if it's easier to walk through the EasyMesh cell question, the N+1 proposal, and the mobility timeline live.

Best regards,  
NK